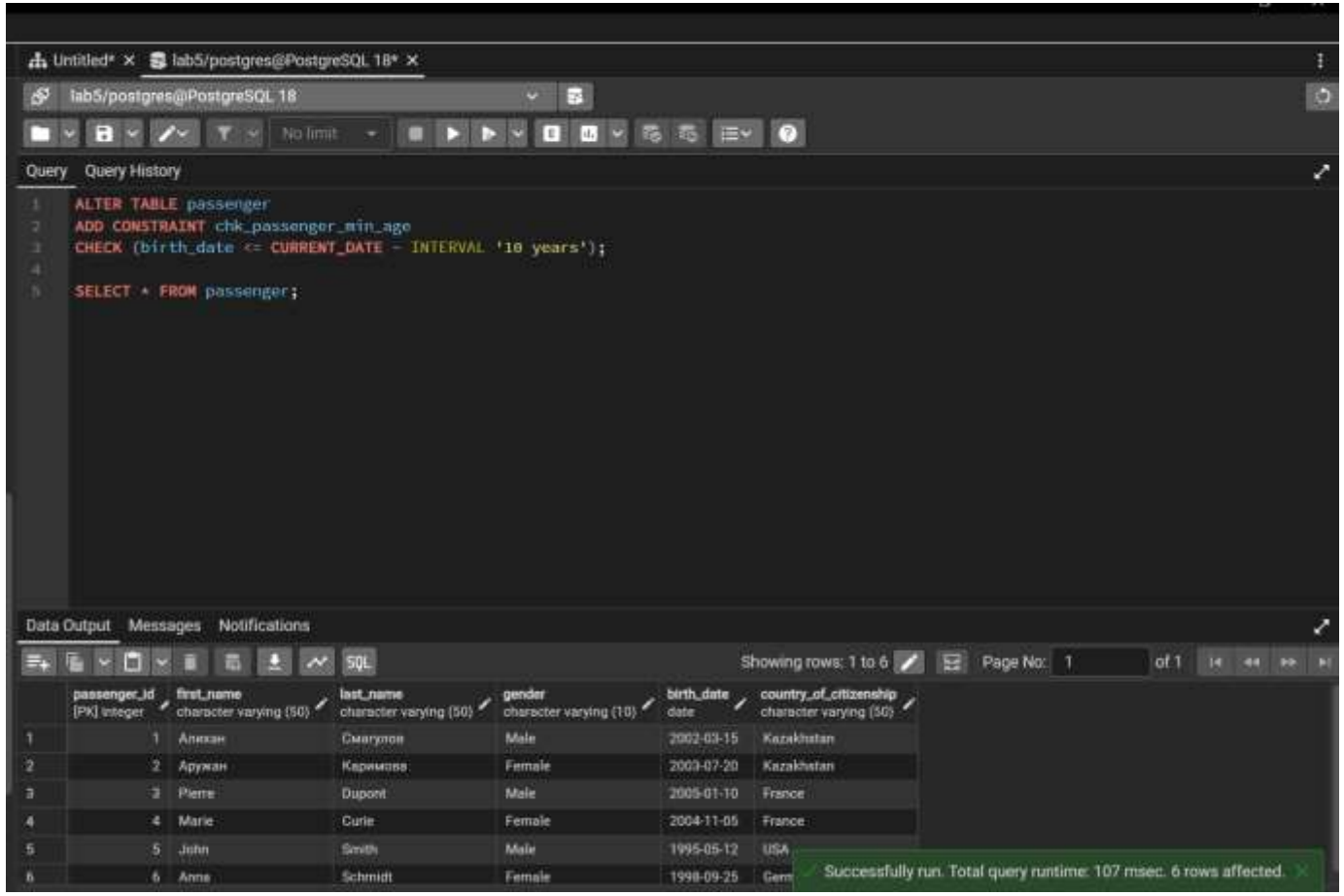
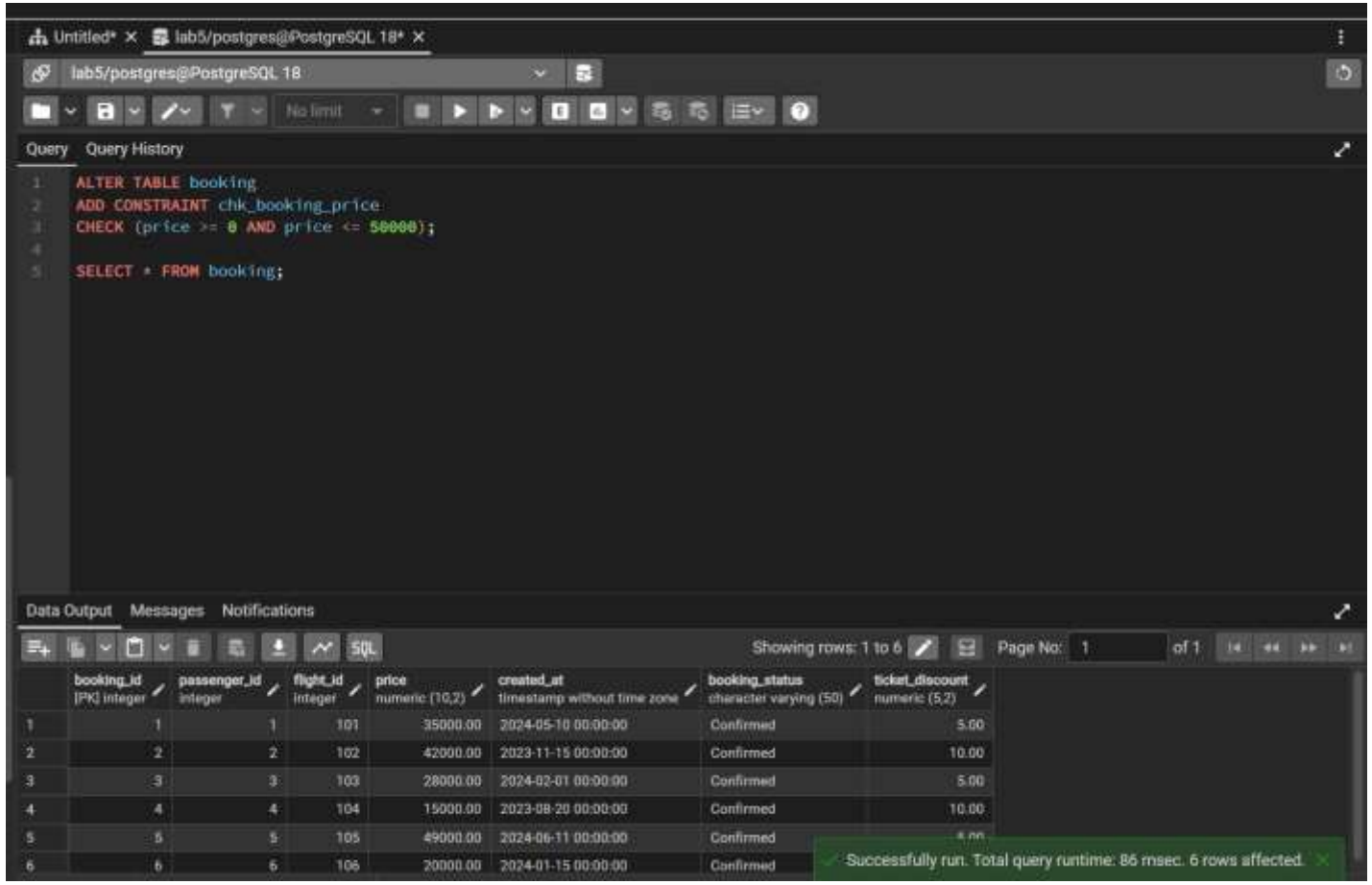
Lab5 data base

Kairzhan Khamza

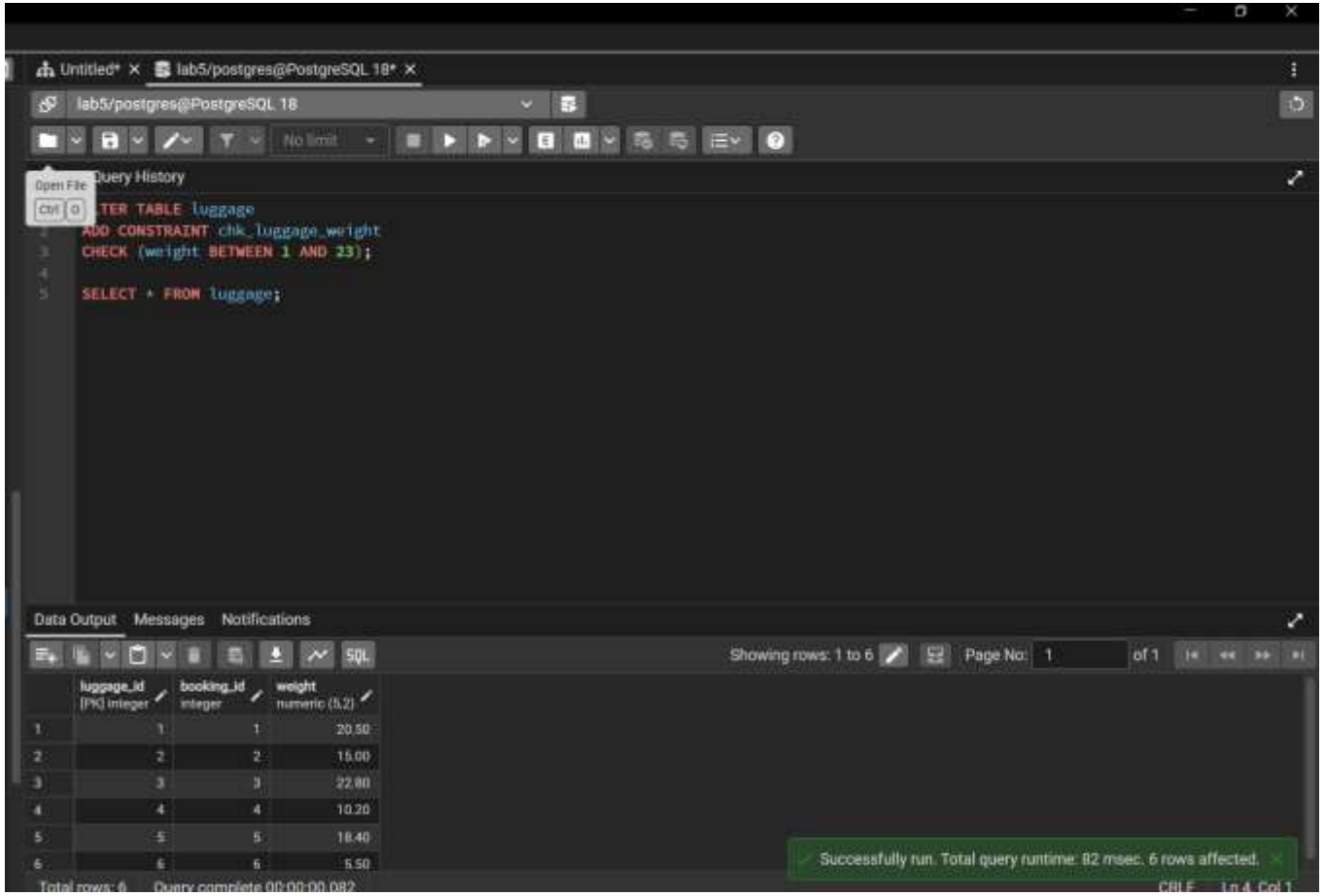
Task 1


Untitled*
lab5/postgres@PostgreSQL 18*
lab5/postgres@PostgreSQL 18
No limit
Query
Query History
ALTER TABLE passenger
ADD CONSTRAINT chk_passenger_min_age
CHECK (birth_date <= CURRENT_DATE - INTERVAL '10 years');
SELECT * FROM passenger;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 6
Page No: 1
of 1
passenger_id [PK] integer
first_name character varying (50)
last_name character varying (50)
gender character varying (10)
birth_date date
country_of_citizenship character varying (50)
1 1 Алихан Смагулов Male 2002-03-15 Kazakhstan
2 2 Аружан Каримова Female 2003-07-20 Kazakhstan
3 3 Pierre Dupont Male 2005-01-10 France
4 4 Marie Curie Female 2004-11-05 France
5 5 John Smith Male 1995-05-12 USA
6 6 Anna Schmidt Female 1998-09-25
Successfully run. Total query runtime: 107 msec. 6 rows affected.

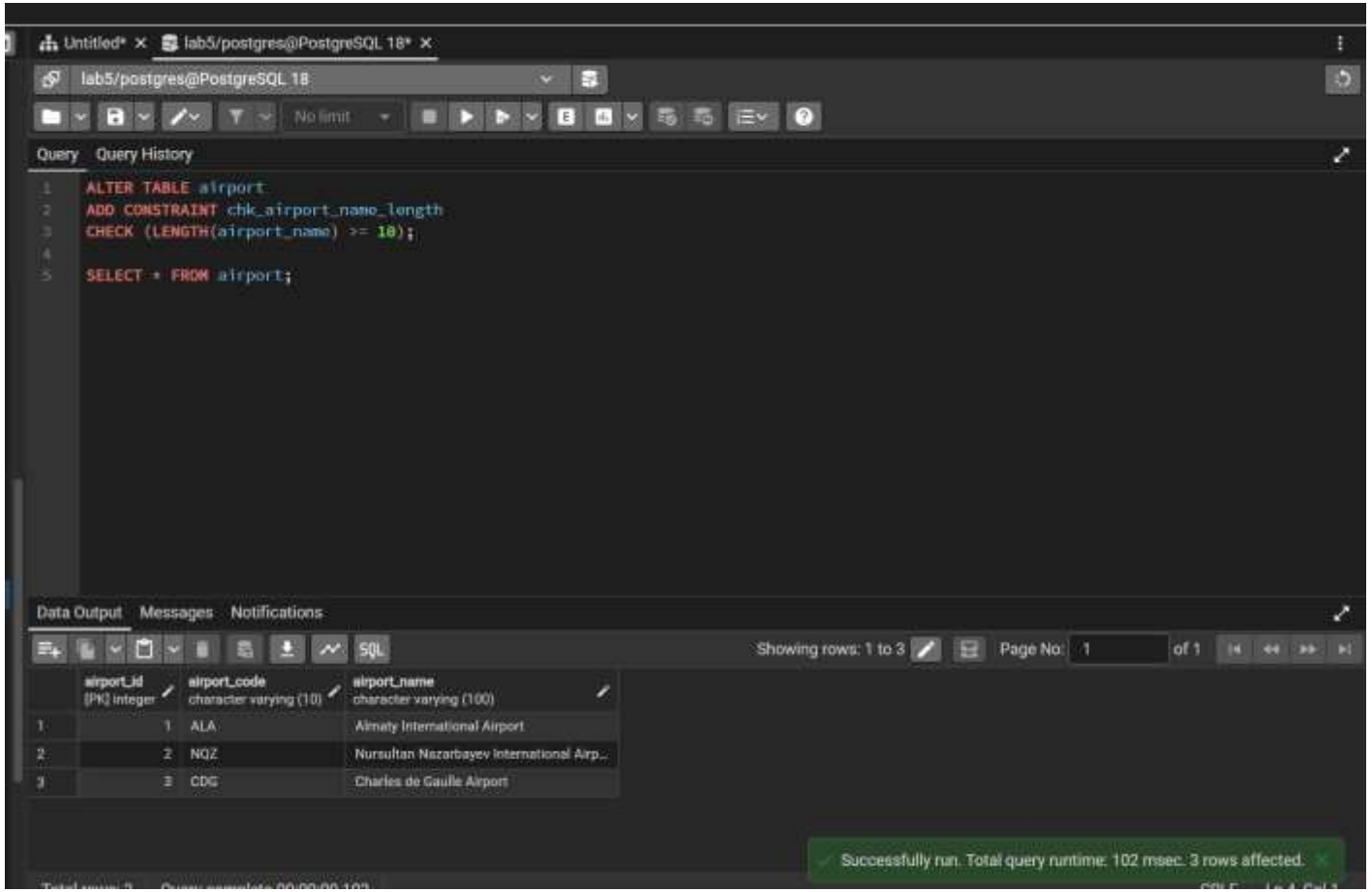
Task 2


Untitled*
lab5/postgres@PostgreSQL 18*
lab5/postgres@PostgreSQL 18
No limit
Query
Query History
ALTER TABLE booking
ADD CONSTRAINT chk_booking_price
CHECK (price >= 0 AND price <= 50000);
SELECT * FROM booking;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 6
Page No: 1 of 1
booking_id [PK] integer
passenger_id integer
flight_id integer
price numeric (10,2)
created_at timestamp without time zone
booking_status character varying (50)
ticket_discount numeric (5,2)
1 1 1 101 35000.00 2024-05-10 00:00:00 Confirmed 5.00
2 2 2 102 42000.00 2023-11-15 00:00:00 Confirmed 10.00
3 3 3 103 28000.00 2024-02-01 00:00:00 Confirmed 5.00
4 4 4 104 15000.00 2023-08-20 00:00:00 Confirmed 10.00
5 5 5 105 49000.00 2024-06-11 00:00:00 Confirmed
6 6 6 106 20000.00 2024-01-15 00:00:00 Confirmed
Successfully run. Total query runtime: 86 msec. 6 rows affected.

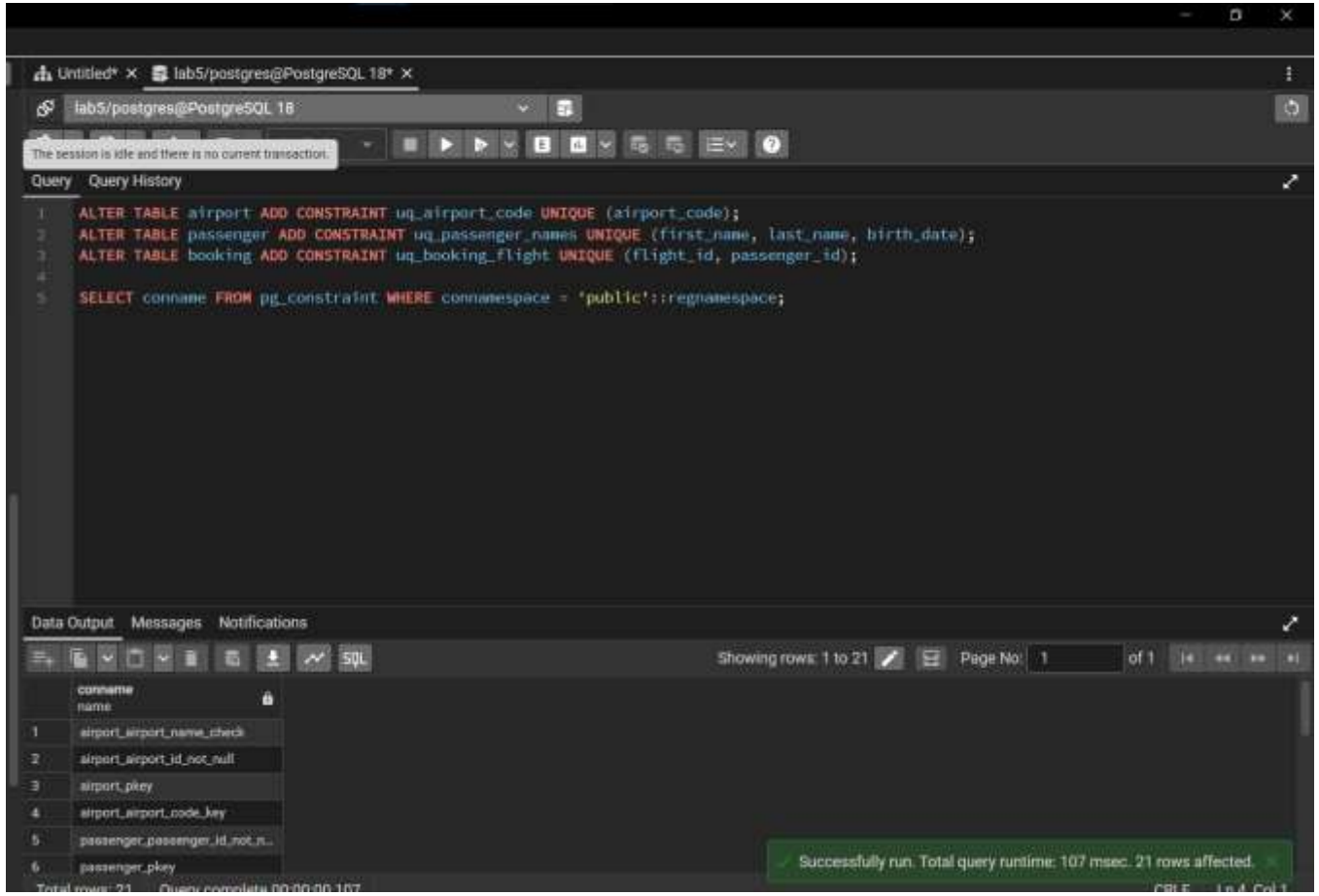
Task 3


Untitled*
lab5/postgres@PostgreSQL 18*
lab5/postgres@PostgreSQL 18
Query History
Open File
Ctrl O
TER TABLE luggage
ADD CONSTRAINT chk_luggage_weight
CHECK (weight BETWEEN 1 AND 23);
SELECT * FROM luggage;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 6
Page No: 1
of 1
luggage_id (PK) integer
booking_id integer
weight numeric (5,2)
1 1 1 20.50
2 2 2 15.00
3 3 3 22.80
4 4 4 10.20
5 5 5 18.40
6 6 6 5.50
Successfully run. Total query runtime: 82 msec. 6 rows affected.
Total rows: 6
Query complete 00:00:00.082
CRLF
Ln 4, Col 1

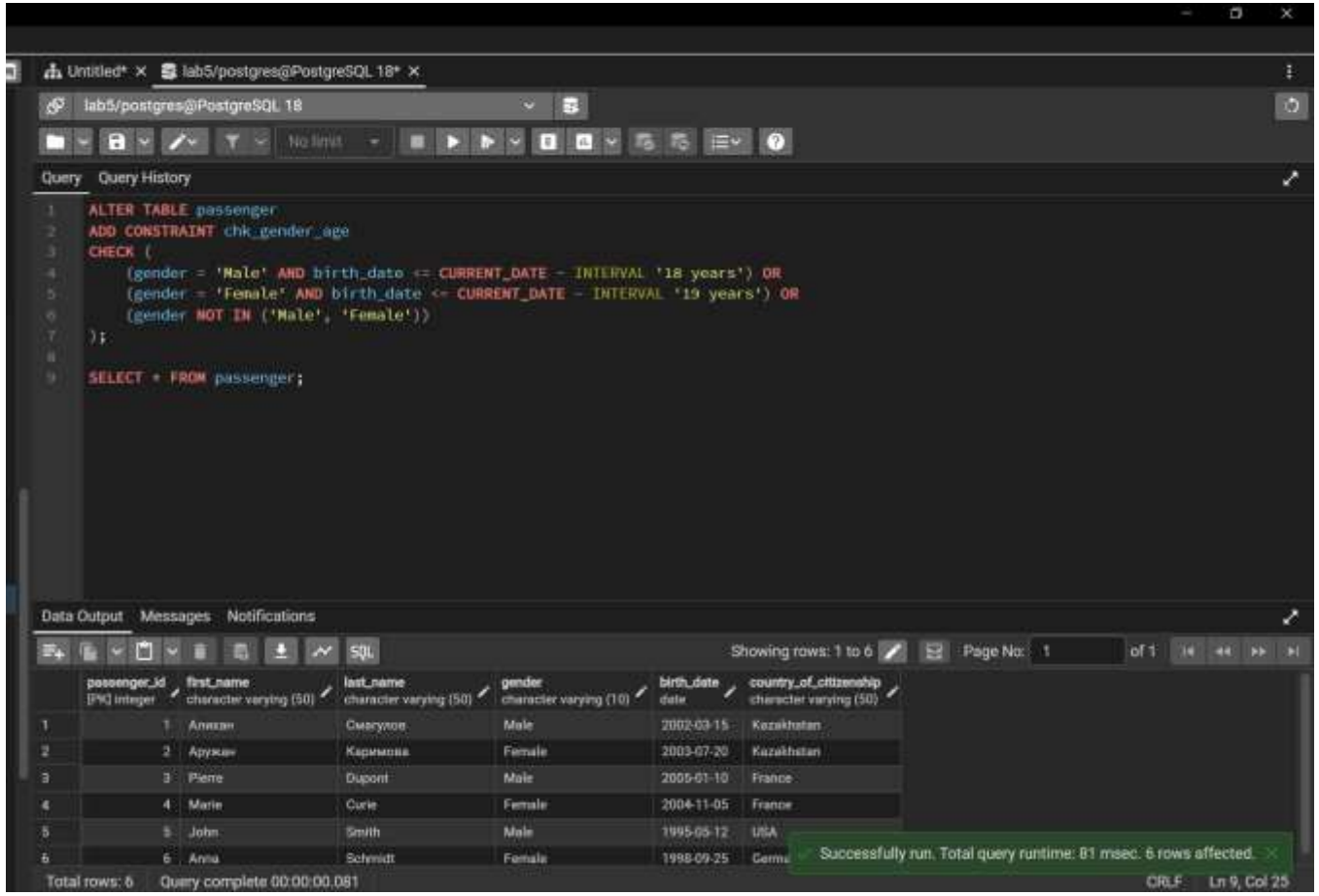
Task 4


Untitled*
lab5/postgres@PostgreSQL 18*
lab5/postgres@PostgreSQL 18
No limit
Query
Query History
ALTER TABLE airport
ADD CONSTRAINT chk_airport_name_length
CHECK (LENGTH(airport_name) >= 10);
SELECT * FROM airport;
Data Output
Messages
Notifications
Showing rows: 1 to 3
Page No: 1 of 1
airport_id [PK] integer
airport_code character varying (10)
airport_name character varying (100)
1 1 ALA Almaty International Airport
2 2 NQZ Nursultan Nazarbayev International Airp...
3 3 CDG Charles de Gaulle Airport
Successfully run. Total query runtime: 102 msec. 3 rows affected.

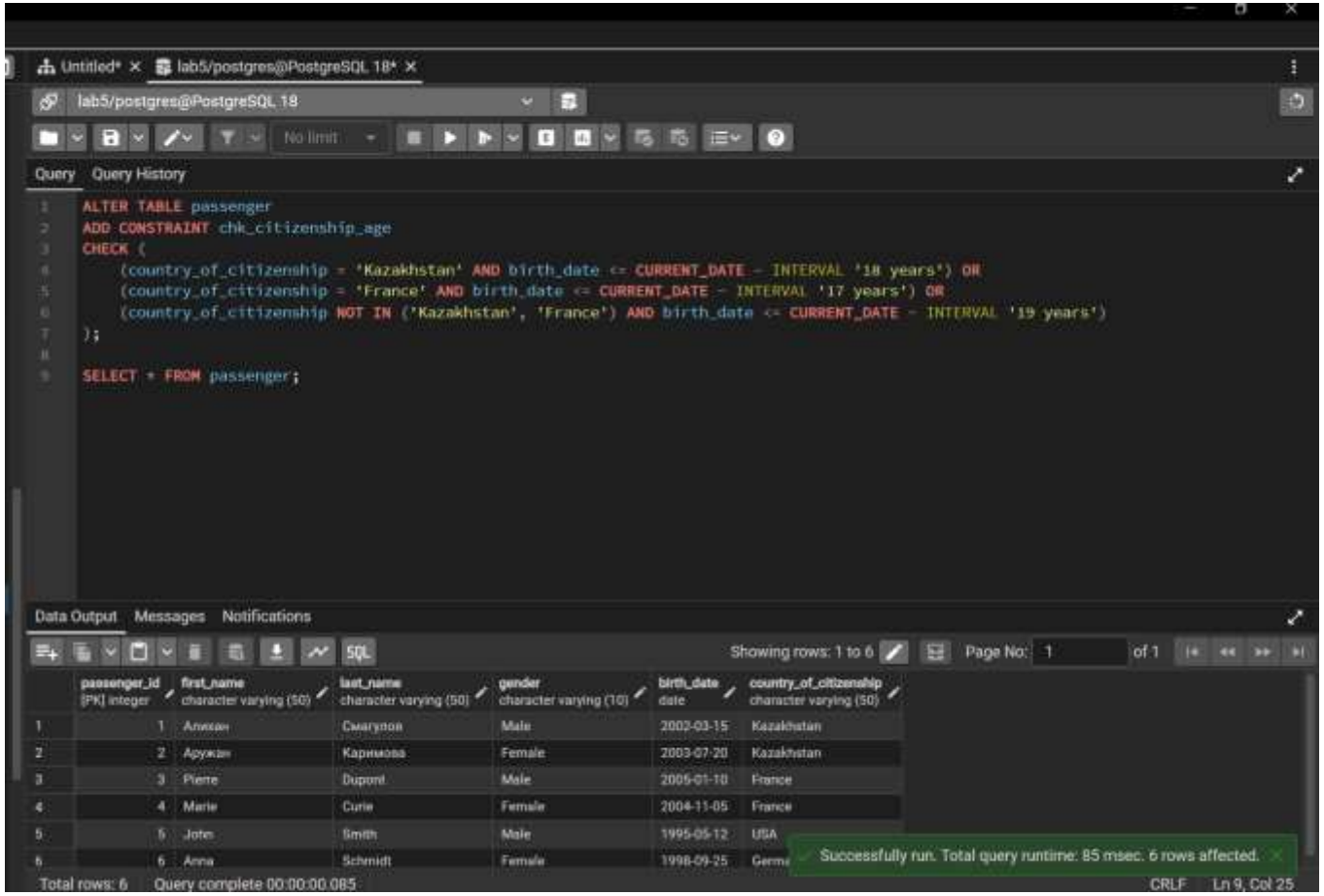
Task 5


Untitled*
lab5/postgres@PostgreSQL 18*
lab5/postgres@PostgreSQL 16
The session is idle and there is no current transaction.
Query
Query History
1 ALTER TABLE airport ADD CONSTRAINT uq_airport_code UNIQUE (airport_code);
2 ALTER TABLE passenger ADD CONSTRAINT uq_passenger_names UNIQUE (first_name, last_name, birth_date);
3 ALTER TABLE booking ADD CONSTRAINT uq_booking_flight UNIQUE (flight_id, passenger_id);
4
5 SELECT conname FROM pg_constraint WHERE connamespace = 'public'::regnamespace;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 21
Page No: 1
of 1
conname
name
1 airport_airport_name_check
2 airport_airport_id_not_null
3 airport_pkey
4 airport_airport_code_key
5 passenger_passenger_id_not_n...
6 passenger_pkey
Successfully run. Total query runtime: 107 msec. 21 rows affected.

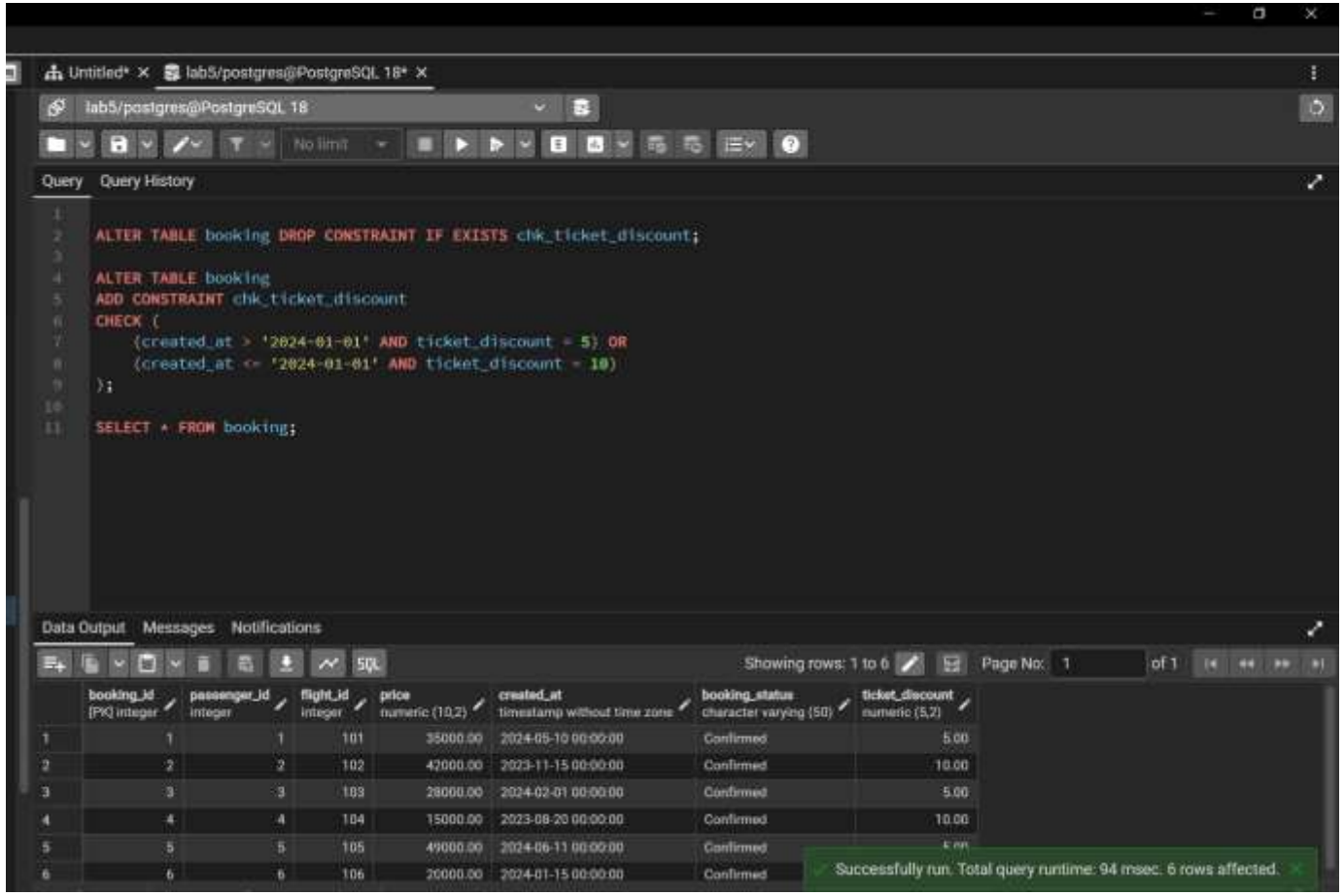
Task 6


Untitled*
lab5/postgres@PostgreSQL 18*
lab5/postgres@PostgreSQL 18
Query
Query History
ALTER TABLE passenger
ADD CONSTRAINT chk_gender_age
CHECK (
(gender = 'Male' AND birth_date <= CURRENT_DATE - INTERVAL '18 years') OR
(gender = 'Female' AND birth_date <= CURRENT_DATE - INTERVAL '19 years') OR
(gender NOT IN ('Male', 'Female'))
);
SELECT * FROM passenger;
Data Output
Messages
Notifications
Showing rows: 1 to 6
Page No: 1
of 1
passenger_id [PK] integer
first_name character varying (50)
last_name character varying (50)
gender character varying (10)
birth_date date
country_of_citizenship character varying (50)
1 1 Алихан Смагулов Male 2002-03-15 Kazakhstan
2 2 Аружан Каримова Female 2003-07-20 Kazakhstan
3 3 Pierre Dupont Male 2005-01-10 France
4 4 Marie Curie Female 2004-11-05 France
5 5 John Smith Male 1995-05-12 USA
6 6 Anna Schmidt Female 1998-09-25
Successfully run. Total query runtime: 81 msec. 6 rows affected.
Total rows: 6
Query complete 00:00:00.081
CRLF
Ln 9, Col 25

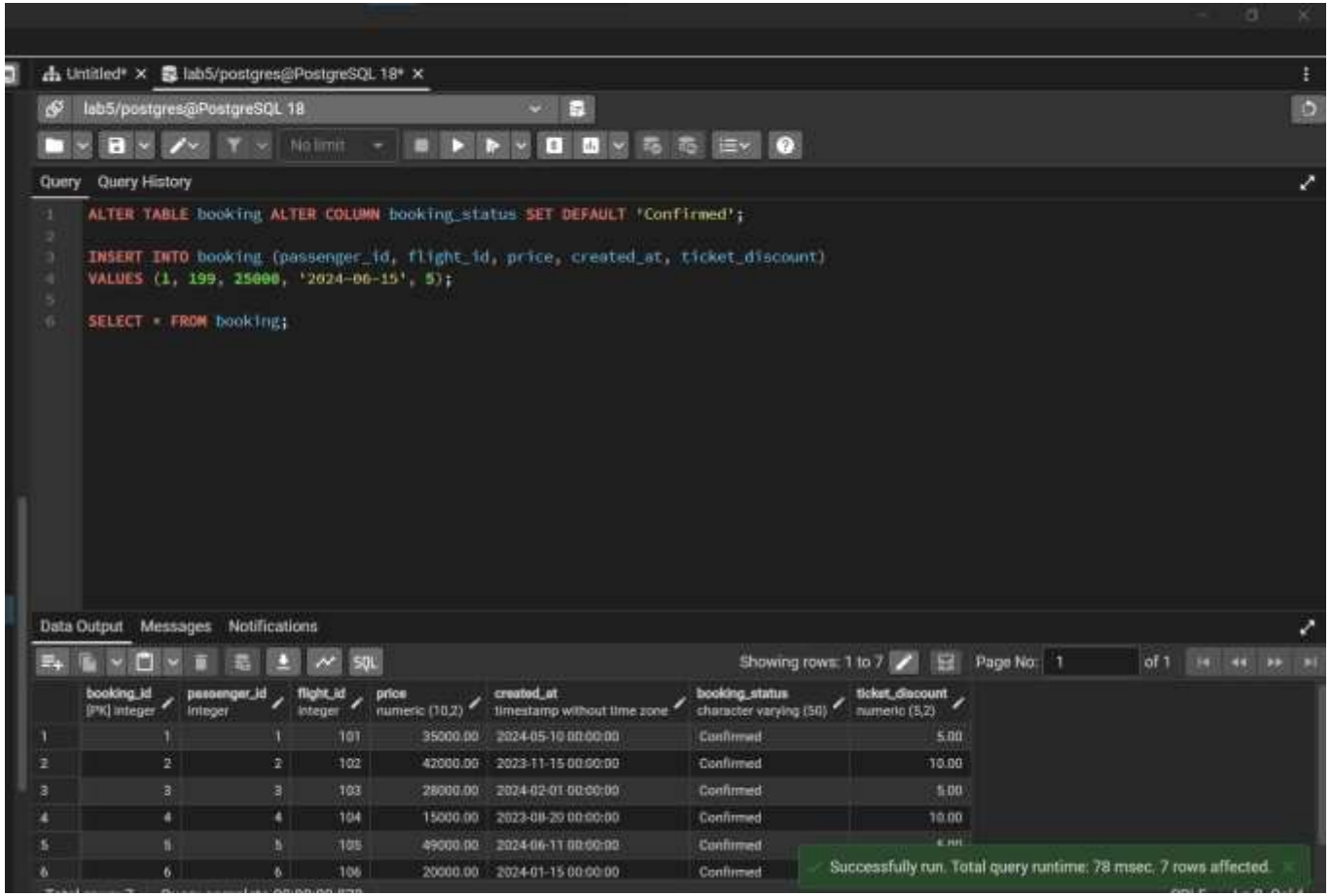
Task 7


Untitled*
lab5/postgres@PostgreSQL 18*
lab5/postgres@PostgreSQL 18
No limit
Query
Query History
ALTER TABLE passenger
ADD CONSTRAINT chk_citizenship_age
CHECK (
(country_of_citizenship = 'Kazakhstan' AND birth_date <= CURRENT_DATE - INTERVAL '18 years') OR
(country_of_citizenship = 'France' AND birth_date <= CURRENT_DATE - INTERVAL '17 years') OR
(country_of_citizenship NOT IN ('Kazakhstan', 'France') AND birth_date <= CURRENT_DATE - INTERVAL '19 years')
);
SELECT * FROM passenger;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 6
Page No: 1
of 1
passenger_id [PK] integer
first_name character varying (50)
last_name character varying (50)
gender character varying (10)
birth_date date
country_of_citizenship character varying (50)
1 1 Алихан Сыгаупов Male 2002-03-15 Kazakhstan
2 2 Аружан Каримова Female 2003-07-20 Kazakhstan
3 3 Pierre Dupont Male 2005-01-10 France
4 4 Marie Curie Female 2004-11-05 France
5 5 John Smith Male 1995-05-12 USA
6 6 Anna Schmidt Female 1998-09-25
Successfully run. Total query runtime: 85 msec. 6 rows affected.
Total rows: 6
Query complete 00:00:00.085
CRLF
Ln 9, Col 25

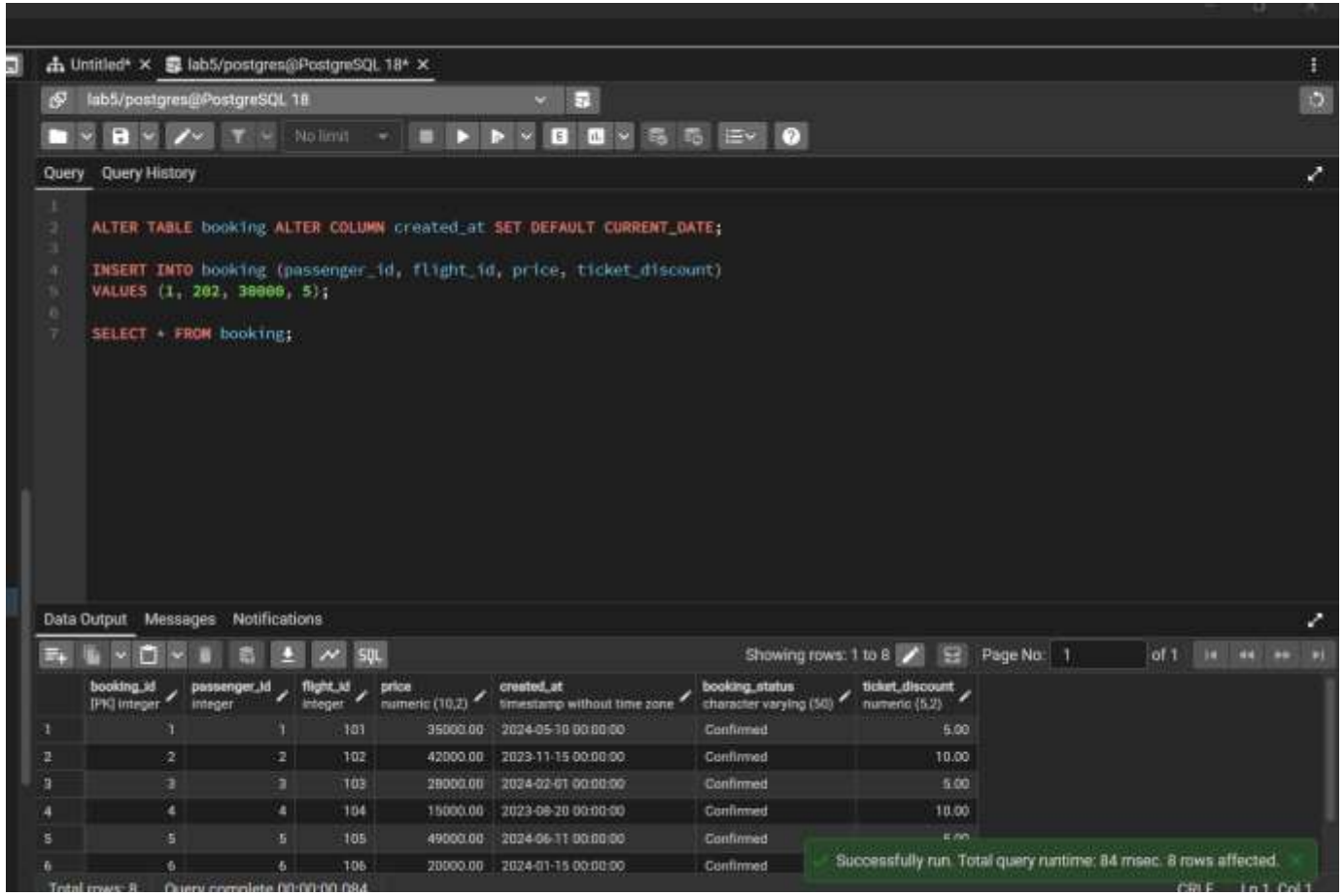
Task 8


Untitled*
lab5/postgres@PostgreSQL 18*
lab5/postgres@PostgreSQL 18
No limit
Query
Query History
ALTER TABLE booking DROP CONSTRAINT IF EXISTS chk_ticket_discount;
ALTER TABLE booking
ADD CONSTRAINT chk_ticket_discount
CHECK (
(created_at > '2024-01-01' AND ticket_discount = 5) OR
(created_at <= '2024-01-01' AND ticket_discount = 10)
);
SELECT * FROM booking;
Data Output
Messages
Notifications
Showing rows: 1 to 6
Page No: 1
of 1
booking_id (PK) integer
passenger_id integer
flight_id integer
price numeric (10,2)
created_at timestamp without time zone
booking_status character varying (50)
ticket_discount numeric (5,2)
1 1 1 101 35000.00 2024-05-10 00:00:00 Confirmed 5.00
2 2 2 102 42000.00 2023-11-15 00:00:00 Confirmed 10.00
3 3 3 103 28000.00 2024-02-01 00:00:00 Confirmed 5.00
4 4 4 104 15000.00 2023-08-20 00:00:00 Confirmed 10.00
5 5 5 105 49000.00 2024-06-11 00:00:00 Confirmed
6 6 6 106 20000.00 2024-01-15 00:00:00 Confirmed
Successfully run. Total query runtime: 94 msec. 6 rows affected.

Task 9


Untitled*
lab5/postgres@PostgreSQL 18*
lab5/postgres@PostgreSQL 18
No limit
Query
Query History
ALTER TABLE booking ALTER COLUMN booking_status SET DEFAULT 'Confirmed';
INSERT INTO booking (passenger_id, flight_id, price, created_at, ticket_discount)
VALUES (1, 199, 25000, '2024-06-15', 5);
SELECT * FROM booking;
Data Output
Messages
Notifications
Showing rows: 1 to 7
Page No: 1
of 1
booking_id [PK] integer
passenger_id integer
flight_id integer
price numeric (10,2)
created_at timestamp without time zone
booking_status character varying (50)
ticket_discount numeric (5,2)
1 1 1 101 35000.00 2024-05-10 00:00:00 Confirmed 5.00
2 2 2 102 42000.00 2023-11-15 00:00:00 Confirmed 10.00
3 3 3 103 28000.00 2024-02-01 00:00:00 Confirmed 5.00
4 4 4 104 15000.00 2023-08-20 00:00:00 Confirmed 10.00
5 5 5 105 49000.00 2024-06-11 00:00:00 Confirmed
6 6 6 106 20000.00 2024-01-15 00:00:00 Confirmed
Successfully run. Total query runtime: 78 msec. 7 rows affected.

Task 10


Untitled*
lab5/postgres@PostgreSQL 18*
lab5/postgres@PostgreSQL 18
No limit
Query
Query History
ALTER TABLE booking ALTER COLUMN created_at SET DEFAULT CURRENT_DATE;
INSERT INTO booking (passenger_id, flight_id, price, ticket_discount)
VALUES (1, 202, 30000, 5);
SELECT * FROM booking;
Data Output
Messages
Notifications
Showing rows: 1 to 8
Page No: 1
of 1
booking_id [PK] integer
passenger_id integer
flight_id integer
price numeric (10,2)
created_at timestamp without time zone
booking_status character varying (50)
ticket_discount numeric (5,2)
1 1 1 101 35000.00 2024-05-10 00:00:00 Confirmed 5.00
2 2 2 102 42000.00 2023-11-15 00:00:00 Confirmed 10.00
3 3 3 103 28000.00 2024-02-01 00:00:00 Confirmed 5.00
4 4 4 104 15000.00 2023-08-20 00:00:00 Confirmed 10.00
5 5 5 105 49000.00 2024-06-11 00:00:00 Confirmed
6 6 6 106 20000.00 2024-01-15 00:00:00 Confirmed
Successfully run. Total query runtime: 84 msec. 8 rows affected.